



CERTIFICATE OF CALIBRATION

BMN MSI Co., Ltd, Lo 32 LK 2 KDT DAI THANH, THANH TRI, HANOI, VIETNAM
Tel: +84-24-730-99-661 <https://www.bmnmed.com>

Environmental Conditions		
Temperature	74 (23.3)	°F (°C)
Relative Humidity	77	%RH
Barometric Pressure	1013 (34304.1)	inHg (hPa)

Model	6510
Serial Number	65101748002

☒ As Left
☐ As Found

☒ In Tolerance
☐ Out of Tolerance

AeroTrak Calibration Bench			Calibration Due Date
Measurement Variable	System ID	Date Last Calibrated	
Reference Remote	72011541003	08/14/2020	08/14/2021
Flow Meter	40401541004	08/14/2020	08/14/2021
Flow Meter	40431539008	08/14/2020	08/14/2021

Particle Standards				
Particle Size	Standard Uncertainty	Standard Deviation	Lot No.	Expiration Date
0.508 µm	0.004 µm	0.0085 µm	201405	08-30-2021
0.702 µm	0.003 µm	0.0049 µm	204048	10-30-2021
0.994 µm	0.008 µm	0.0100 µm	200992	08-30-2021
5.020 µm	0.040 µm	0.0400 µm	198900	06-30-2021

BMN does hereby certify that the calibration performed on the above described instrument meets the requirements of ISO 21501-4. BMN does hereby certify that the above described instrument conforms to the original manufacturer's specification (not applicable to As Found data) and has been calibrated using standards whose accuracies are traceable to the United States National Institute of Standards and Technology (NIST) or has been verified with respect to instrumentation whose accuracy is traceable to NIST, or is derived from accepted values of physical constants.

February 1, 2021

DATE

CALIBRATED



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Size Calibration and Verification of Size Setting			
Nominal Particle Size	Gain Stage	Digital Cutpoint	Expanded Uncertainty
0.5 μm	A	653	3.9%
0.7 μm	A	1403	3.8%
1.0 μm	A	2029	3.9%
5.0 μm	B	2362	3.9%

Counting Efficiency			
Particle Size	Actual	Allowable Range	Pass/Fail
0.508 μm	55%	30-70%	Pass
0.994 μm	99%	90-110%	Pass

Size Resolution			
Particle Size	Measured	Allowable Range	Pass/Fail
0.508 μm	4.1%	$\leq 15\%$	Pass

False Count Rate						
Sample Time (min)	Sampled (L)	Measured Counts (#)	Concentration ($\#/m^3$)	95% UCL ($\#/m^3$)	Allowable Range ($\#/m^3$)	Pass/Fail
45	1274	3	2.36	1.4	≤ 7.1	Pass

Sampling Flow Rate (L/min) †		
Actual	Allowable Range	Pass/Fail
27.8	27.5-28.0	Pass

Sampling Time Accuracy †		
Measured	Allowable Range	Pass/Fail
$< \pm 0.1\%$	$\pm 1\%$	Pass

Response Rate †		
Measured	Allowable Range	Pass/Fail
0.0006%	$\leq 0.5\%$	Pass

Maximum Particle Concentration †	
29000000 $\#/m^3$ @10% Coincidence Loss	

† Tested and verified during product development

Calibration Interval	
Calibration Date	Expiration Date
February 1, 2021	February 1, 2022